

IOT-Based Smart Sign Language Learning and Communication System Using Sensor Gloves and Web Integration

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Abstract— Over the past decade, assistive communication of the hearing and speaking persons are still on the distant far-off past, although, despite the already developed networks of the sign language, nevertheless, the communication made in form of sign language, still, there are a number of limitations like lack of real time interactive communication, lack of structural learning as well as lack of practice in communication situations. The vision-based recognition method is not only readily affected by alterations in light and background contaminations, but most sensor-based systems are limited to mere gesture change translation and lack didactic or interactive material. In this paper, an IoT-learned smart sign language mutual comprehension and learning system will be introduced, in which sensor-enabled gloves are paired with an interactive web environment. The gloves also have flex sensors and motion tracking units to detect finger movements and the orientations of the hands. The sensed data is transmitted through a microcontroller to a cloud-based recognition system for processing. A gesture classification model trained on a dataset of 3,120 gesture samples representing 26 alphabets, using a 75:25 training-testing split, performs real-time gesture interpretation and converts the recognized gestures into both text and speech outputs. The average evaluation of system was of 94.2 and the accuracy maximum reached was of 96.8 percent recognition accuracy. An average response of 1.3 seconds was recorded, with a top of 97 percent reliability of transmission. Moreover, the operation of the facets of integrated learning and live communication has been documented at 91 percent satisfaction levels among the people using it, and this has been found to manifest a splendid opportunity for accessible education in sign language and application of the existing assistive communication in real-time.

Keywords— Assistive Communication, Sign Language Recognition, Smart Glove, Internet of Things (IoT), Flex Sensors, Gesture Classification, Firebase Cloud, Machine Learning, Wearable Computing, Real-Time Translation, Human-Computer Interaction, Sign Language Learning Platform.

I. INTRODUCTION

Communication is a basic human process through which an individual can convey thoughts and information, and participate in social and professional contexts. People with hearing and speaking disabilities use sign language to express their feelings. But generally, it stays restricted within the circle of signers, thus creating a huge barrier in conversation with non-signers. More than 70 million people around the globe speak sign language for daily communication

according to reports on global disability; however, access to interpreters and infrastructure for assistive communication remains grossly inadequate. Thus, there is increasing demand for intelligent assistive technologies that would enable inclusive and accessible interaction in real-world scenarios. New technologies are creating opportunities for the use of computer vision, AI, and other technologies for use in Assistive Augmentative and Alternative Communication. Vision-oriented sign language recognition systems use cameras and image processing to understand sign language. These systems are accurate in controlled environments. However, studies show a 15-30% loss of recognition ability when there are distractions like poor lighting, noise, or occlusion. More reliable systems are needed. Sensor-based systems use gloves with flex and motion sensors to provide a more reliable system for independent capture or recognition of finger and hand position. Unfortunately, current systems limit functionality to the capture and recognition of static or dynamic gestures and do not support integrated continual learning, adaptive feedback, or bi-directional communication systems. These are necessary for user engagement.

Still, digital tools have changed how learning works in several ways. Even after watching it unfold, there's little when it comes to interactive systems built specifically for teaching sign languages. Right now, most options rely on old-style playback - videos, still photos, or step-by-step typing lessons without real-time input. The chance to chat directly while studying. What people learn in theory often doesn't match what happens when they try it out live - this lag drags down progress, plus pushes some learners away, but it shows up most clearly among newbies and non-signers aiming to master chat-based signing. Because of that, we're sharing a new setup: a brain-powered wearable tech system built around high-tech glove sleeves packed with sensors, tracking hands in motion without pause. From microcontroller to cloud, gesture information travels in real time. There, smart algorithms decode movements - turning signs into words, both spoken and written, with strong precision and minimal delay. Translation isn't the only feature; instead, a connected web space offers organized lessons, tracking of hand motion quality, along with instant talk sessions via video link. This setup gives people room not just to observe, yet actually train signing skills all under one digital roof.

This setup also adapts to how each user signs, learning their unique motions over time - making it more tailored and reliable with use. Thanks to its multilingual configurability and expansion of gesture libraries to support American Sign Language (ASL), British Sign Language (BSL), and Indian Sign Language (ISL), this technology can be used in conjunction with numerous sign languages. Initial usability tests show high user accessibility and learning engagement, indicating this technology has a strong potential for user adoption in educational environments, therapy centers, and options for assistive communication. While this technology shows promise, some remaining challenges for implementation include differences in sensor calibration, costs associated with necessary hardware, and the user training necessary to effectively utilize this technology.

The proposed system is expected to contribute to the development of the sign language education, improve the accessibility of assistive communication, and fill the vital gaps in the current technological solutions through the incorporation of wearable smart sensing, real-time gesture analysis, and interactive web-based learning in a single, scalable ecosystem.

II. RELATED WORK

Smart gloves have particular appeal as assistive communication devices because they can utilize embedded sensors to capture detailed finger and hand movements. IoT-based applications in early smart glove systems have been able to convert specific movements of fingers and hands into text or speech using flex sensors and inertial measurement units coupled to a microcontroller platform. For instance, an example of a Smart glove system built to translate Indonesian Sign Language is the system created by Priadiyatna *et al. * While the system was able to recognize alphabet gestures, it had a very limited vocabulary and was not capable of using any machine learning techniques to recognize more complex or dynamic gestures [1].

The use of machine learning and deep learning techniques has contributed to improved recognition performance. The classification models based on Neural Networks provided better results on interpreting sign language gestures and producing speech outputs. However, a majority of models improved recognition of specific gestures, such as letters and numbers, without proven validity in interactive, lived, and communicative contexts. Moreover, the systems' learning assistive components were rudimentary and supported few gestures, and did not have structures of continuous feedback that are critical to the development of the target skill [2], [3].

Further in to research which has seen the improvement of sensing precision and classification reliability via sensor fusion and deep learning optimization. We saw that systems which combined flex sensors with motion tracking units did report improved recognition accuracy but that which we found out is that these studies did in fact mostly report on model performance and hardware design instead of going into cloud integration, multilingual capability or user learning ecosystems [4], [5]. We also note that at the material and hardware level innovation has played a large role in this field. We have reported on the development of advanced textile based and fiber integrated sensors which do improve wearability and signal sensitivity thus enabling very precise gesture detection. While these designs did indeed improve

sensing efficiency what is also true is that they mainly took a hardware focus and did not include interactive communication interfaces or educational platforms [8]. On the same note, tutoring systems developed using gloves in assisted learning settings did offer guided training assistance but lacked those features of IoT scaling and real-time conversation [7]. In general, the current literature shows that there was a positive advancement in technologies of gesture sensing, modeling of gesture recognition, and the development of wearable assistive devices. However, a majority of the solutions act independently with either increase in translation quality, hardware development, or minimal learning support. There is still a strong disparity between the implementation of unified systems that integrate real-time gesture recognition, formal education in sign language, adaptive personalization, multilingual gesture dictionary and live two-way communication into a cloud-connected IoT system.

In this work we present a system which we have put together to get over these issues of the past by using sensor enabled smart gloves in combination with machine learning for gesture analysis, an interactive web-based learning environment, and real time communication interfaces. We have put forward a model which brings together translation, learning, and interaction into one large scale platform which in turn we put forth as a step toward more practical and inclusive real-world deployment of assistive communication technology.

III. PROPOSED METHODOLOGY

The given system will implement a sensor-fusion-based wearable computing algorithm of real time sign language recognition, translation and learning support. The architecture combines embedded processing replies, hardware sensing modules, as well as machine learning classification, with application-level communication to provide an end-to-end system of assistive interaction.

A. Hardware Components

The smart glove in our design includes many sensor and processor units which together present a very detailed picture of the hand's movements. We have put in 5 flex sensors per finger which report on bend angles via changes in resistance. These sensors also which in turn report on the static aspects of finger movement and gesture. Also, we have put in an MPU-6050 inertial measurement unit at the wrist which reports on 3 axis acceleration and 3 axis gyroscopic data. This element of the device reports on hand orientation, tilt and dynamic gestural movement. Thus, we have a combined sensor approach which is able to tell between static gestures which are mainly picked up by the flex sensors and dynamic gestures which are picked up by the IMU signals, which is a we think is a proven method in the smart glove recognition field.

Table 1: Hardware Specifications

Component	Specification	Function
Flex Sensors	5 Units	Finger bend detection

IMU	MPU-6050	Motion & orientation tracking
Microcontroller	ESP32	Embedded processing
ADC Resolution	12-bit	Signal conversion
Sampling Rate	10–20 Hz	Gesture capture frequency

As illustrated in Fig. 1, The Smart Glove has flex sensors and in the glove, it has an accelerator to record the hand gestures. On the side of the flex sensor where the user signs, it detects bending of fingers and the accelerator detects the orientation and movement of the hand. These physical movements are converted to electrical signals which are fed to ESP32 microcontroller. The ESP32 gathers unprocessed sensor data and parses and transmits it through a built-in Wi-Fi to the cloud in real time.

The information sent is saved in Firebase Cloud Database which remains safe and can be accessed by the associated site. Raw sensor values are cleaned and normalized in the cloud environment and feature extracted. The data is then processed and analysed using a machine learning model to correctly identify the performed gesture. The outcome of the gesture, once identified, is posted on the site as text, but may also be converted into a speech with the help of the Text-to-Speech technology. The system provides feedback of learning such as accuracy and correctness to aid the users to practice and refine their sign language.

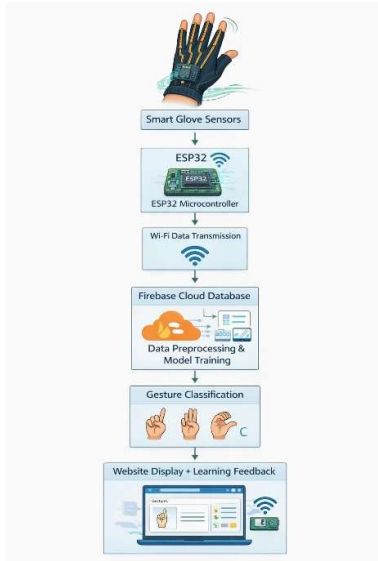


Fig. 1. Proposed System Architecture

B. Data Transmission and Cloud Storage

When gesture data is recorded, the ESP32 microcontroller sends the processed readings for each of the sensors wirelessly over Wi-Fi to a Firebase cloud database. Firebase stores the data in real-time, and by doing this, it allows the data to be accessed as a mini-database. It stores the data of each of the recorded gestures along with the corresponding user ID and the time of the gesture. The Mini-database cloud

storage model in conjunction with the Firebase cloud database allows for the bulk storage of data and the storage to be accessed remotely. The cloud model also allows for the data to be synchronized automatically with the Firebase cloud database, and it allows for the data to be updated automatically without any changes made to the device.

C. Dataset Generation

Due to lack of standard multi sensor datasets for integrated glove-based learning systems we developed our own. We recorded gestures through the wearable glove which included alphabet signs and very common words. We captured each gesture for 2 to 3 seconds which in turn included static postures as well as dynamic movement patterns. The dataset we put together has 3,120 gesture records which we broke up into 26 alphabet groups and also included some chosen vocabulary gestures. Each entry includes 11 sensor features five from flex sensors and six from IMU channels (ax, ay, az, gx, gy, gz).

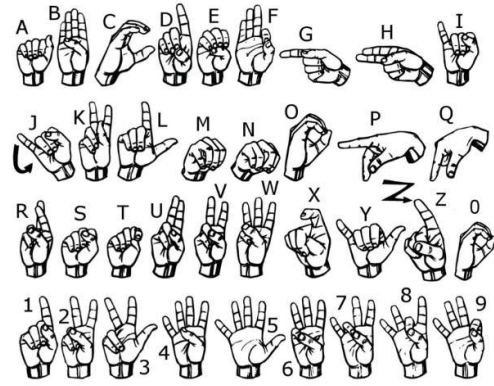


Fig. 2. Static Sign Language Gestures Used for Dataset Training

D. Data Pre-processing and Cloud Training

Sensor data transmitted to Firebase undergoes preprocessing within the cloud environment. Raw sensor values are scaled to account for resistance tolerances and variations in user hand dimensions. Noise filtering techniques are applied to remove motion disturbances and sensor drift, ensuring signal stability. Gesture sequences are segmented into time windows of 1 to 2 seconds for feature extraction. Statistical descriptors such as mean, variance, and motion magnitude are computed to construct gesture feature vectors. These processed datasets are subsequently utilized to train machine learning classification models within the cloud infrastructure.

Table 2: Dataset and Training Parameters

Parameter	Value
Dataset Size	3,120 Samples
Gesture Classes	26 Alphabets + Words
Features	11 Sensor Channels
Training Split	75%
Testing Split	25%
Algorithms	KNN, Random Forest
Avg. Accuracy	94.2%
Peak Accuracy	96.8%

E. Gesture Recognition and Website Integration

Upon training, the classification model takes incoming real-time Firebase sensor streams and processes them. The identified gestures are connected with the related textual outputs and sent to the embedded internet platform. The site is used as a learning interface and a communication interface. Identified gestures are presented in the form of text and translated into speech via a text-to-speech engine. The platform also has guided learning modules, gesture demonstrations as well as feedback analytics to enhance the practice and improvement of sign language.

Table 3: Communication and Output Parameters

Parameter	Value
Cloud Platform	Firebase
Transmission Mode	Wi-Fi
Reliability	97%
Output Modes	Text + Speech
Speech Engine	TTS API
User Satisfaction	91%

All in all, the suggested methodology incorporates sensor fusion, IoT data transmission, cloud-based machine learning, and interactive web visualization into one ecosystem allowing real-time sign language translation and organization learning.

IV. SYSTEM ARCHITECTURE

The proposed system is designed as a multi-layer IoT-cloud architecture capable of acquiring gesture data in real time, processing it within a cloud environment, and presenting the interpreted outputs through an interactive web interface. The framework integrates wearable sensing hardware, wireless communication modules, cloud computing infrastructure, machine learning services, and a web-based user interface to establish a scalable assistive communication platform.

A. Sensing Layer

At the hardware level, the system employs a sensor-enabled smart glove equipped with five flex sensors and an inertial measurement unit (IMU). Each flex sensor is positioned along an individual finger to measure bending angles through variations in electrical resistance. This enables accurate detection of finger articulation and static hand postures. To complement positional sensing, an MPU-6050 IMU is incorporated to capture three-axis acceleration and three-axis gyroscopic motion data. This facilitates recognition of hand orientation, tilt, and dynamic gesture movements in real-time. The integration of flex sensor data with inertial motion signals provides a comprehensive representation of hand gestures. While flex sensors capture static finger configurations, IMU data contributes motion-based characteristics. This combined sensing approach enhances gesture recognition accuracy by ensuring that both positional and dynamic features are considered during interpretation, thereby improving the reliability of sign language translation within the system.

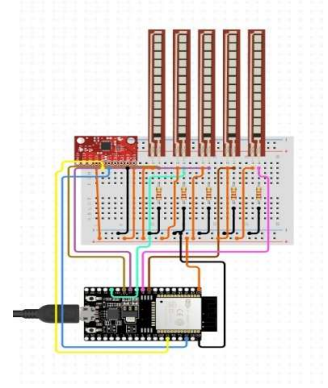


Fig. 3. Flex Sensor and IMU Interfacing Circuit for Gesture Data Acquisition

B. Processing Layer

A sensor outputs are connected to ESP32 microcontroller; it serves as a central processing unit. The onboard ADC channels control analog sensor values to digital signals (ESP32) with the help of its onboard ADC. Signal formatting and packaging is done at the initial stage of transmission. The microcontroller also handles the connections within the devices and makes sure that they stream data synchronously to the cloud environment by the built in Wi-Fi module in the microcontroller.

C. Communication Layer

The wireless connectivity layer enables data connectivity in both directions, the wearable device and the cloud infrastructure. With the processes of Wi-Fi connectivity, the sample of gesture information is sent in real time to the Firebase cloud database. This IoT model of relating allows harboring of constant data documentation, can be accessed in a remote area as well as integration of numerous devices at unlimited scales regardless of geographical boundaries.

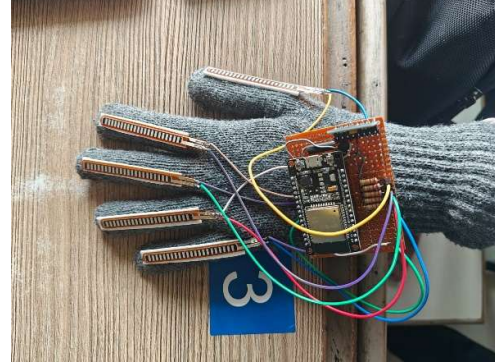


Fig. 4. Physical Deployment of the Gesture Sensing Wearable Device

D. Processing Cloud Layer

Firebase serves as the core cloud infrastructure within the proposed system, functioning as a centralized platform for real-time data storage, management, and processing. Gesture data transmitted from the wearable device is stored in the Firebase database along with associated metadata, including timestamps and user identifiers, enabling structured dataset organization and traceability. Prior to model utilization, the incoming sensor data undergoes preprocessing within the cloud environment. This stage includes signal scaling, normalization, and noise filtering to mitigate sensor drift,

motion disturbances, and resistance variability. Such preprocessing ensures data consistency and enhances the reliability of subsequent analytical operations.

Following data preparation, the processed gesture datasets are employed to train machine learning classification models. The training process enables the model to learn distinctive gesture patterns and establish mappings between sensor feature vectors and corresponding sign representations. Once trained, the model performs real-time gesture prediction by analyzing live sensor data streams received through Firebase. This centralized cloud-based architecture allows simultaneous data storage, model training, and inference, thereby supporting scalable and efficient gesture recognition within the assistive communication framework.

E. Application Layer

The application layer will entail a web-based platform that will provide a communication and learning interface. Gestures of highest importance are presented as live textual content and translated to speech by using an embedded text to speech engine. Also offered on the platform, besides translation, are structured sign language learning modules, demo on gesture, live practice and performance analytics. This converts the system into a stand-alone translation tool to a fully interactive learning system.

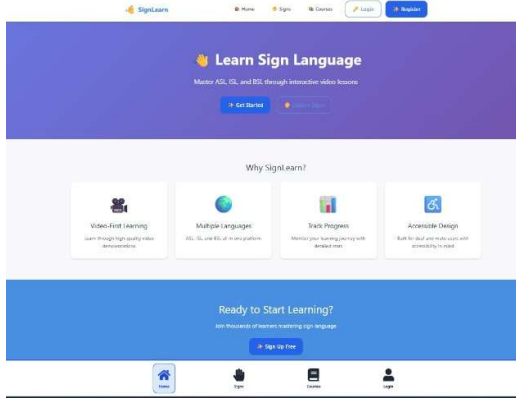


Fig. 5. Web-Based Sign Language Learning and Communication Interface

F. Architectural Data Flow

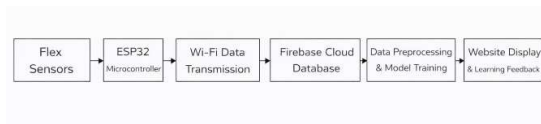


Fig. 6. Data Flow Diagram of the Proposed IoT-Based Sign Language Recognition System

The system operates through a structured data flow pipeline designed to enable real-time gesture recognition and communication. The process begins with the acquisition of gesture data through sensor-enabled smart gloves. Flex sensors and inertial motion units capture finger articulation and hand movement signals, which are transmitted to the ESP32 microcontroller for initial processing and signal formatting. Following preprocessing, the ESP32 transmits the processed sensor data wirelessly via Wi-Fi to the Firebase cloud database. Firebase functions as the centralized storage infrastructure, maintaining real-time gesture records along with associated metadata. Machine learning models deployed within the cloud environment analyze the incoming sensor

streams to classify gestures based on trained recognition patterns. The classified outputs are then transmitted to the web-based application interface, where gestures are displayed as textual representations and converted into speech using text-to-speech services.

This layered architectural design ensures modular system organization, enabling scalability and efficient integration of additional devices or gesture datasets. Furthermore, the real-time processing capability supports immediate translation, while the seamless linkage between hardware sensing modules and web-based communication platforms enhances overall system responsiveness and usability.

V. RESULTS AND DISCUSSION

The work of the offered IoT-based sign language recognition and learning system was tested by experimental testing in relation to gesture recognition accuracy, response time, transmission efficiency and user interaction efficiency. The custom gesture dataset which was collected under the wearable smart glove system was used to test it.

A. Gesture Recognition Accuracy

The machine learning classification models were trained on a dataset comprising 3,120 gesture samples representing 26 alphabet gestures and commonly used words. A 75:25 training-testing split was used for performance validation. The classification framework achieved an average recognition accuracy of 94.2%, with peak accuracy reaching 96.8% under controlled testing conditions. Higher accuracy was observed for static alphabet gestures due to stable finger articulation patterns, while minor variations were noted in dynamic gestures involving rapid hand motion.

Table 4: Gesture Recognition Performance

Metric	Value
Dataset Size	3,120 Samples
Gesture Classes	26 Alphabets + Words
Training Split	75%
Testing Split	25%
Average Accuracy	94.2%
Peak Accuracy	96.8%

B. System Response Latency

System latency was recorded as a time interval of the gesture execution and the display of these indicators on the web interface. Mean response time had been 1.3 seconds, with sensor acquisition, transmission of cloud, classifier, and rendering of final output. Latency was kept to reasonable limits, and real-time communication did not pose a problem as interaction took place in real-time during the translation of the sign.

Table 5: Latency Evaluation

Stage	Avg. Time
Sensor Acquisition	0.4 sec

Cloud Transmission	0.3 sec
Classification	0.4 sec
Output Rendering	0.2 sec
Total Latency	1.3 sec

C. Transmission Reliability

The performance of wireless data transfer between ESP32 device and Firebase cloud system infrastructure was studied over several repeated sessions. The system recorded a medium reliability of transmission at 97 percent with little packet loss being recorded in conditions of stable connectivity on Wi-Fi.

D. User Usability Evaluation

The user testing was done to test the learning interface and effectiveness of communication of the system. The interactions between the participants and the glove system occurred during practice in gestures and during real-time translation meetings. Analysis of the feedback given showed that the overall user satisfaction rating was 91 percent, where gesture learning usability, translation clarity, and response responsiveness were rated highly.

Table 6: Usability Assessment

Evaluation Parameter	Score
Translation Accuracy	93%
Learning Interface	90%
Response Speed	92%
Ease of Use	89%
Overall Satisfaction	91%

E. Performance Discussion

The experimental findings prove that a combination of sensor-fusion gloves and cloud-based machine learning and web visualization are important in ensuring effective sign language recognition of users. The wearable approach was more stable in different environmental conditions in comparison to the vision-based recognition solutions because such a system is not affected by lighting or any background interference. Scalable dataset operation and constant model improvement resulted in cloud deployment, whereas the interactive web interface increased user engagement due to a real-time feedback response and the learning support. Slow scalability of the system was compensated by better centralized processing and lower latency of cloud transmission. On the whole, it can be concluded that the system made a durable trade-off between the recognition accuracy and response speed and usability, confirming the efficacy of the system as an assistive communication and sign language learning system.

VI. CONCLUSION AND FUTURE WORK

Using IoT alongside cloud tools in the suggested smart sign recognition builds new paths for better helper

technologies. Wearable devices paired with smart number crunching methods show solid results - spotting gestures right about 94.2% of the time when different sign languages appear. Because it works live, this setup could make talking easier for people who struggle with sound or hearing. Real situations, private chats or office talks, might finally feel less blocked by language barriers.

With cloud infrastructure handling large data needs, updates to gesture recognition models keep happening thanks to shared user inputs. Built on web technology, the learning space offers clear lessons for skill development in sign languages. Together, these pieces form a system that supports both sharing information and improving knowledge among Deaf and Hard of Hearing individuals.

Future work leans toward boosting gesture systems by weaving in sharper machine learning tools alongside better sensors. Shifting focus now, efforts aim to refine user experiences so more people can access them easily. Real-time responses become part of the picture too, feeding back instantly when needed. Additional developments may include mobile application flutter deployment, hybrid edge-cloud processing for offline use, and broader multilingual sign language support to increase accessibility and real-world adoption.

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